

IMS Hands-On Lab

z/OS Connect and IMS OpenAPI 2

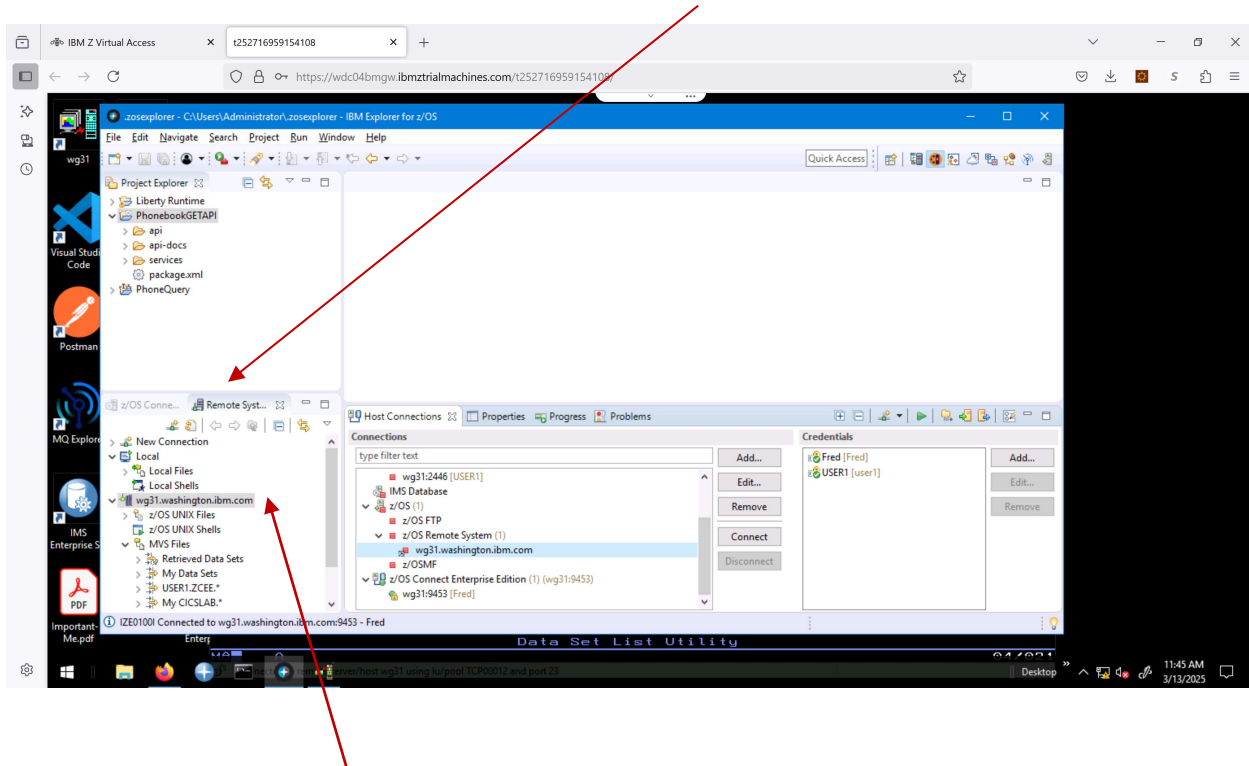
Troubleshooting

When problems occur, or if you need a closer look at the interactions that are occurring, there are several places which provide information,

- | | |
|---|--------------|
| 1. z/OS Connect environment traces/messages log | p. 2 |
| 2. Chec the z/OS Connect server in z/OS for errors | p. 7 |
| 3. Check the IMS control region for errors | p. 8 |
| 4. Check IMS Connect for errors | p. 9 |
| 5. Turn on the IMS Connect recorder trace | p. 10 |

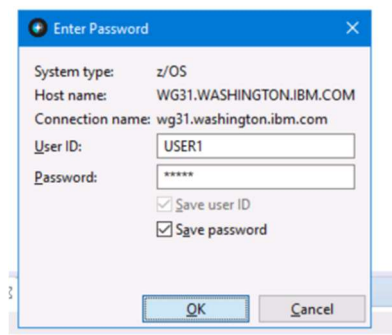
1. For the z/OS Connect environment, traces and message logs:

1-a. In z/OS Explorer, click on the Remote Server tab

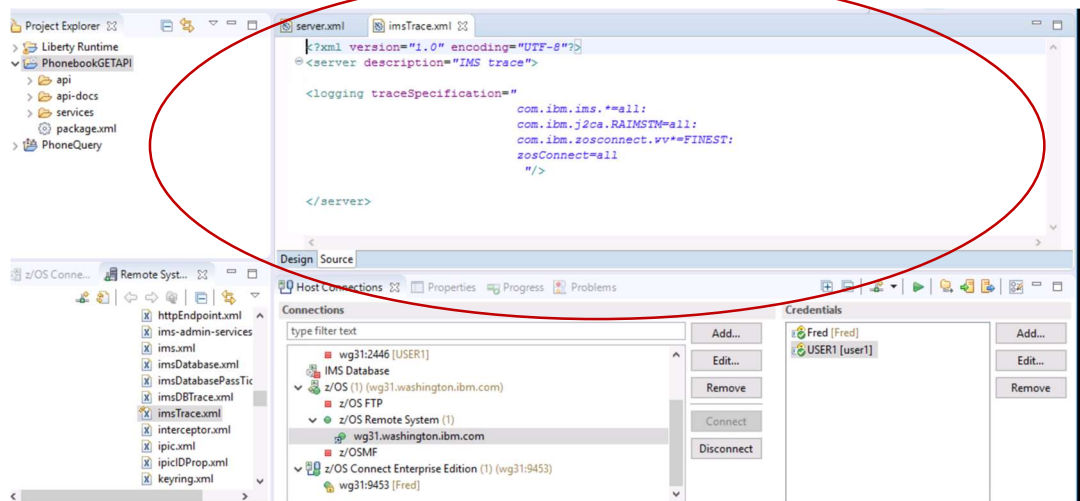


Expand wg31.washington.ibm.com

1-b. When challenged for userid/password, key in USER1 / USER1

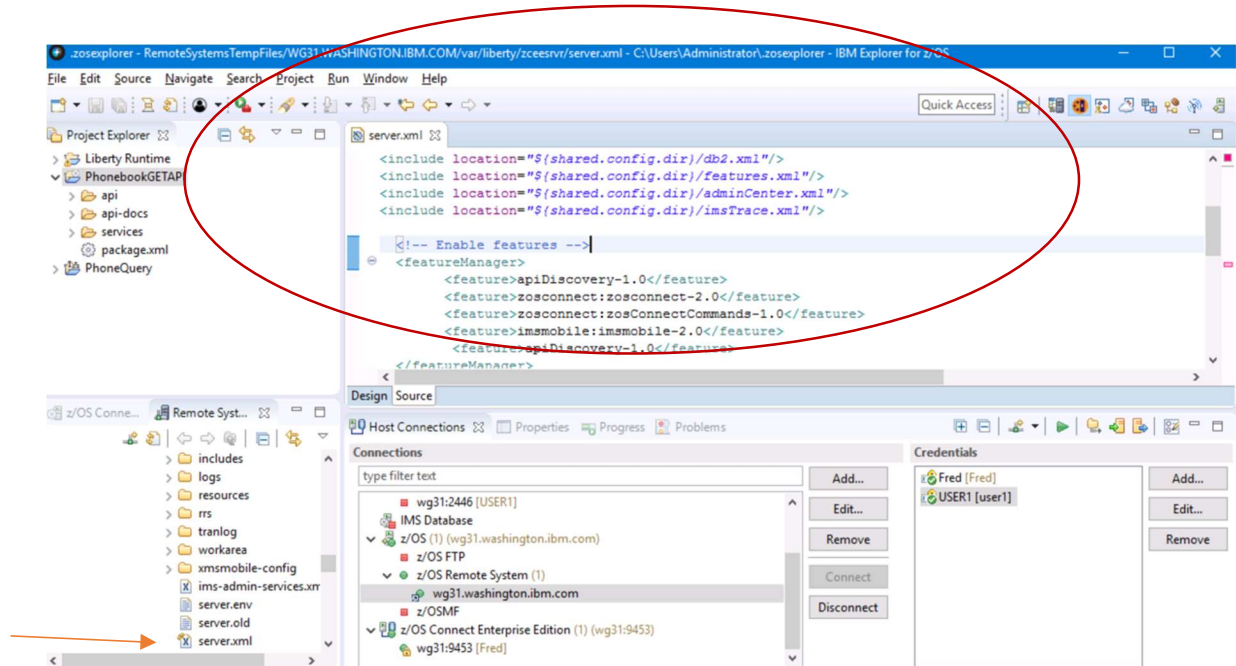


1-c. Follow the path: **z/OS Unix File > / > var /Liberty > zceesrvr** in the **includes > config** --- look at imsTraces.xml



1-d. This specification will need to be included in the server.xml file.

If this line does not exist in the server.xml file, add it in and save.

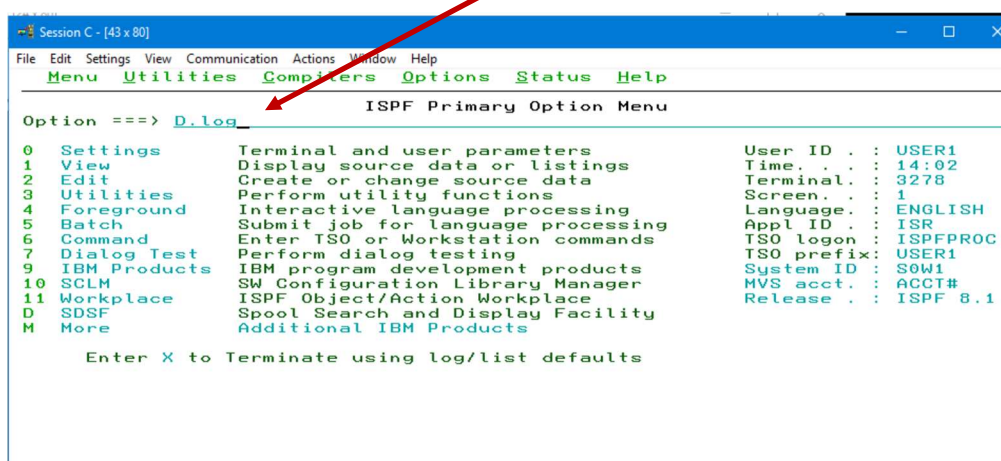


If the line was already there, then the trace is on, continue to the directions in 1-e,

Otherwise, make the change, save the server.xml file.

If you had to add the line, you will need to refresh the z/OS Connect server on the z/OS system.

- Log on to **TSO** using the USER1/USER1 id and password
 - On the **ISPF** panel, key in **D.LOG**



```
Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Menu Utilities Computers Options Status Help

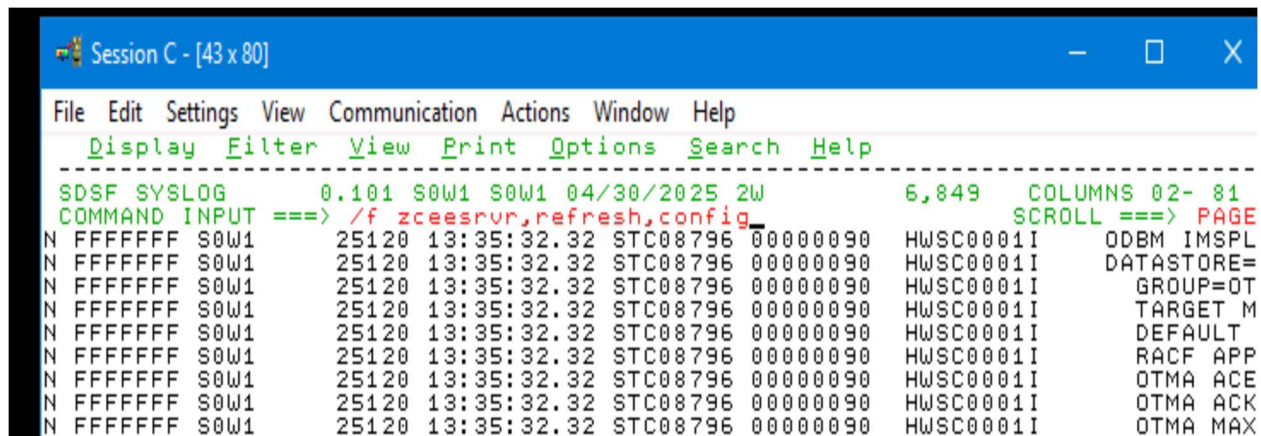
Option ==> D.log

0 Settings      Terminal and user parameters
1 View         Display source data or listings
2 Edit         Create or change source data
3 Utilities     Perform utility functions
4 Foreground   Interactive language processing
5 Batch        Submit job for language processing
6 Command      Enter TSO or Workstation commands
7 Dialog Test  Perform dialog testing
9 IBM Products IBM program development products
10 SCLM        SW Configuration Library Manager
11 Workplace   ISPF Object/Action Workplace
D SDSF        Spool Search and Display Facility
M More         Additional IBM Products

User ID . : USER1
Time . . : 14:02
Terminal . : 3278
Screen . . : 1
Language . : ENGLISH
Appl ID . : ISR
TSO logon : ISPFPROC
TSO prefix: USER1
System ID : S0W1
MVS acct. : ACCT#
Release . : ISPF 8.1

Enter X to Terminate using log/list defaults
```

- This will open up SDSF SYSLOG panel where you can issue the command to refresh the ZCEESRVR
- Key in: **/f zceesrvr,refresh,config**



```
Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help

SDSF SYSLOG 0.101 S0W1 S0W1 04/30/2025 2W 6,849 COLUMNS 02- 81
COMMAND INPUT ==> /f zceesrvr,refresh,config SCROLL ==> PAGE

N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I ODBM IMSPL
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I DATASTORE=
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I GROUP=OT
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I TARGET M
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I DEFAULT
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I RACF APP
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I OTMA ACE
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I OTMA ACK
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC00001I OTMA MAX
```

- To View the server messages in the job... on the command input line, key in: **pre zceesrvr**

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG 0.101 S0W1 S0W1 04/30/2025 2W 6,849 COLUMNS 02- 81
COMMAND INPUT ==> Pre zceesrvr SCROLL ==> PAGE
RESPONSE=S0W1
+CWJKB00051: COMMAND RESPONSES COMPLETED SUCCESSFULLY FROM File
Refresh Command Handler.
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I TARGET ME
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I DEFAULT R
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I RACF APPL
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I OTMA ACEE
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I OTMA ACK
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I OTMA MAX
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I SUPER MEM
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I IMSPLEX=PLE
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I MEMBER=IM
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I ODBM=IVP150
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I AI TDS=TRP

```

- And then an **st** to see the jobs that match the prefix you entered.

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG 0.101 S0W1 S0W1 04/30/2025 2W 6,849 COLUMNS 02- 81
COMMAND INPUT ==> st SCROLL ==> PAGE
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I ODBM IMSPLE
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I DATASTORE=I
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I GROUP=OTM
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I TARGET ME
N FFFFFFFF S0W1 25120 13:35:32.32 STC08796 00000090 HWSC0001I DEFAULT R

```

- Select the active server

```

File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF STATUS DISPLAY ALL CLASSES LINE 1-1 (1)
COMMAND INPUT ==> SCROLL ==> PAGE
NP JOBNAME JobID Owner Prty Queue C Pos Saff ASys Status
S_ ZCEESRVR STC08754 ATSSERV 15 EXECUTION S0W1 S0W1

```

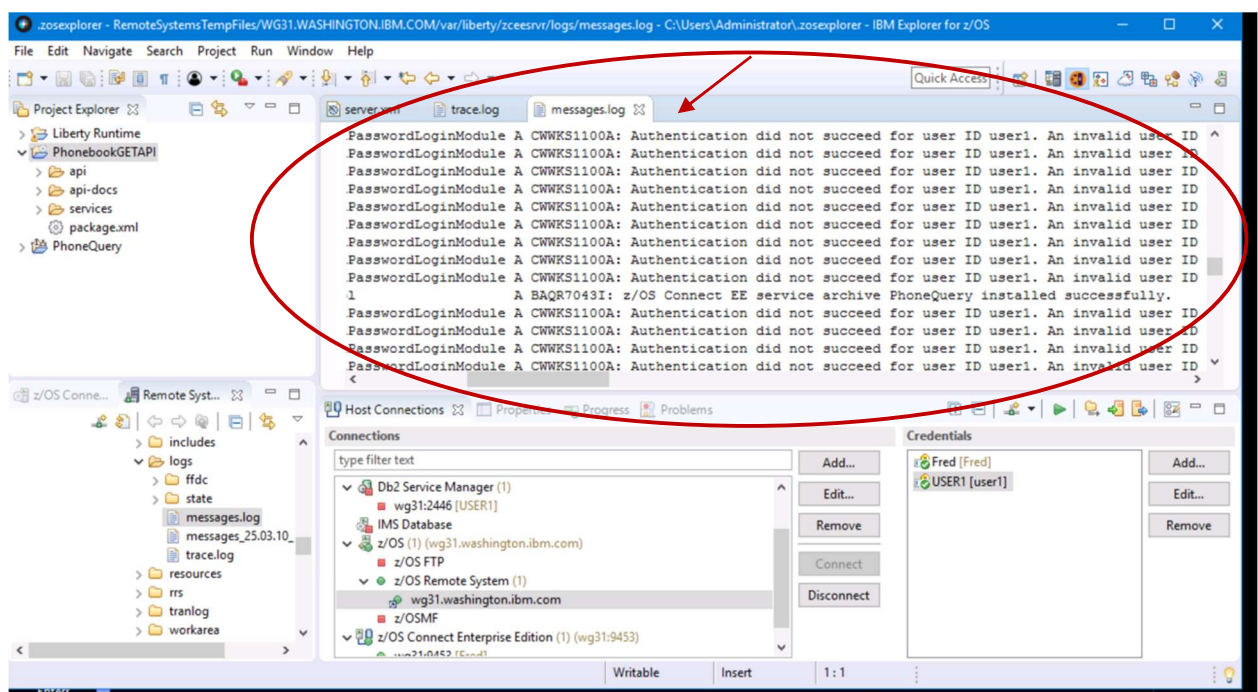
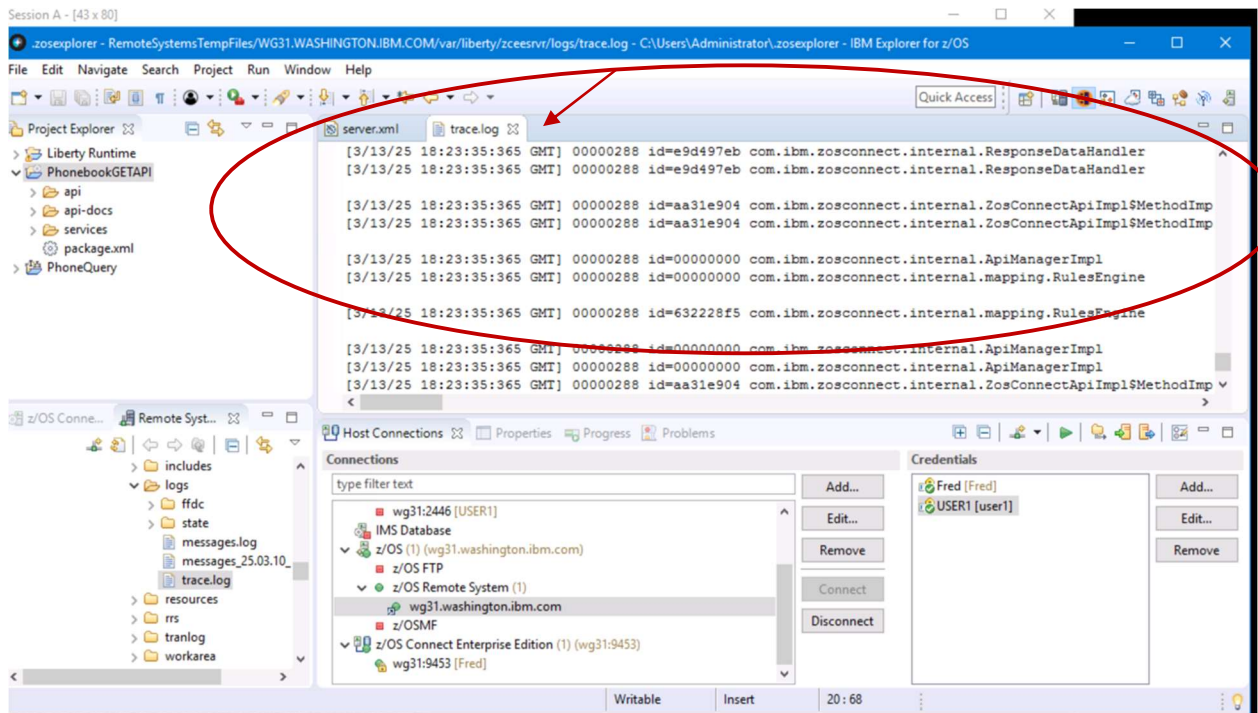
- At the bottom, you should see that the server configuration was updated

```

WKG0028A: Processing included configuration resource: /var/ats/zosconnect/server
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/server
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
WKG0028A: Processing included configuration resource: /var/ats/zosconnect/shared
AS0018I: The trace state has been changed. The new trace state is *=info:com.ibm
AS0040I: The configured trace state included the following specifications that
WKG0017I: The server configuration was successfully updated in 3.591 seconds.
***** BOTTOM OF DATA *****

```

- 1- e. Once the server is available with trace specifications, then while the system is running both messages and trace information can be reviewed.



2. You can also check the messages on the z/OS Connect server running in z/OS.

- Log on to TSO and on the ISPF panel, select **D** for SDSF.
- Once in the SDSF panel, key in **pre zceesrvr** to set the prefix of jobs you want to see, then key **ST**.

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF MENU 3.1      ADCDPL      S0W1      LINE 1-38 (98)
COMMAND INPUT ==> pre ZCEESRVRR_      SCROLL ==> PAGE
NP  NAME          Description      Group      Status
   DA      Active users      Jobs
   I      Input queue      Jobs
   O      Output queue      Output
   H      Held output queue      Output
   ST      Status of jobs      Jobs
   JG      Job groups      JES
   SYM      System symbols      System
   LOG      System log      Log
   SR      System requests      Log
   MAS      Members in the MAS      JES
   JC      Job classes      JES
   SE      Scheduling environments      WLM
   RES      WLM resources      WLM
   FNC      Forcloses      WLM

```

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF MENU 3.1      ADCDPL      S0W1      LINE 1-38 (98)
COMMAND INPUT ==> ST_      SCROLL ==> PAGE
NP  NAME          Description      Group      Status
   DA      Active users      Jobs
   I      Input queue      Jobs
   O      Output queue      Output
   H      Held output queue      Output
   ST      Status of jobs      Jobs
   JG      Job groups      JES

```

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF STATUS DISPLAY ALL CLASSES      LINE 1-1 (1)
COMMAND INPUT ==>      SCROLL ==> PAI
NP  JOBNAME      JobID      Owner      Prty Queue      C      Pos SAff ASys Status
S_  ZCEESRVRR  STC08754  ATSSERV      15 EXECUTION      C      S0W1 S0W1

```

- Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of the address space you want to review. In an active system, there may be many address spaces so the above directions get you to a specific address space.
- You can view all the different messages that might indicate an error condition

```

Y5/1/25 15:02:11:264 GMT" 00000406 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:02:11:294 GMT" 0000040a y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:02:11:327 GMT" 00000447 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:02:11:370 GMT" 00000406 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:11:02:057 GMT" 00000447 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:11:05:137 GMT" 00000401 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:11:05:486 GMT" 00000401 com.ibm.zosconnect.internal.ApiManagerImpl
Y5/1/25 15:11:05:541 GMT" 00000401 com.ibm.zosconnect.internal.ApiManagerImpl
Y5/1/25 15:11:05:907 GMT" 0000040a y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:11:05:936 GMT" 0000040a y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:12:45:250 GMT" 00000406 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:12:45:477 GMT" 00000406 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:15:02:217 GMT" 00000401 com.ibm.ws.logging.internal.impl.IncidentImpl
Y5/1/25 15:15:02:250 GMT" 00000447 com.ibm.ws.logging.internal.impl.IncidentImpl
Y5/1/25 15:16:32:466 GMT" 0000040a com.ibm.j2ca.RAIMSTM
Y5/1/25 15:17:15:686 GMT" 00000401 y.authentication.jaas.modules.UsernameAndPass
Y5/1/25 15:17:15:700 GMT" 00000401 com.ibm.j2ca.RAIMSTM
***** BOTTOM OF DATA *****

```

3. To view the IMS Control region for error messages or to review the environment

- Once again, in TSO – on the ISPF panel, select **D** for SDSF.
- Key in: **pre IMS15CR1** on the command input line
- Key in: ST
- Select the IMS15CR1 server
 - Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of IMS15CR1

```

.      REGID JOBNAME      TYPE  TRAN/STEP PROGRAM  STATUS      CLASS LINE 001 P1
      1 IMSMSG      TP      NONE LINE 001 PTERM 001.      1 LINE 001 PTERM
      JMPRGN      JMP      NONE LINE 001 PTERM 001.
      JBPRGN      JBP      NONE LINE 001 PTERM 001.
      BATCHREG      BMP      NONE LINE 001 PTERM 001.
      FPRGN      FP      NONE LINE 001 PTERM 001.
      2 IMS15ODM      DBT      IMS15ODM      AVAILABLE      LI
      IMS15RC1      DBRC      LINE 001 PTERM 001.
      IMS15DL1      DLS      LINE 001 PTERM 001.
      VTAM ACB OPEN      -LOGONS ENABLED      LINE 001 PTERM 001.
      IMSLU=N/A.N/A      APPC STATUS=DISABLED TIMEOUT= 0 MAXC= 5000 LINE
      OTMA GROUP=OTMAGRP      STATUS=ACTIVE      LINE 001 PTERM 001.
      APPC/OTMA SHARED QUEUE STATUS - LOCAL=INACTIVE      GLOBAL=INACTIVE      L
      APPC/OTMA SHARED QUEUES LOGGING=N LINE 001 PTERM 001.
      APPC/OTMA RRS MAX TCBS - 40 ATTACHED TCBS - 2 QUEUED RRSWKS- 0 LI
      APPLID=IMS15      GRSNAME=      STATUS=DISABLED      LINE 001 PTE
      TCP/IP.GENIMSID=      STATUS=DISABLED LINE 001 PTERM 001.
      LINE ACTIVE-IN - 1 ACTIV-OUT - 0 LINE 001 PTERM 001.
      NODE ACTIVE-IN - 0 ACTIV-OUT - 0 LINE 001 PTERM 001.
      *25120/132634*      LINE 001 PTERM 001.
      DFS2500I DATABASE IVPDB1I      SUCCESSFULLY ALLOCATED .
      DFS2500I DATABASE IVPDB1      SUCCESSFULLY ALLOCATED .
      /RCL NODE TCP00007 USER USER1 .
      .DFS058I 14.02.12 RCLSDST COMMAND COMPLETED NODE TCP00007 USER USER1 .
      DFS2864I EXTERNAL TRACE DATASET DFSTRA01 FULL - SWITCHING TO DFSTRA02 ..
      DFS2864I EXTERNAL TRACE DATASET DFSTRA02 FULL - SWITCHING TO DFSTRA01 ..
      DFS2864I EXTERNAL TRACE DATASET DFSTRA01 FULL - SWITCHING TO DFSTRA02 ..
      ***** BOTTOM OF DATA *****

```

4. You can also view the IMS Connect for error messages or to review the environment

- If not already in SDSF, then on TSO – on the ISPF panel, select **D** for SDSF.
- Key in: **pre IMS15HWS** on the command input line
- Key in: ST
- Select the IMS15HWS1 server
 - Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of IMS15HWS

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF OUTPUT DISPLAY IMS15HWS STC08796 DSID 4 LINE 21 COLUMNS 02- 81
COMMAND INPUT ==> SCROLL ==> PAGE
HWSS0780I TCPIP COMMUNICATION ON HOSTNAME=TCPIP OPENED; M=SDOT
HWSS0790I LISTENING ON PORT=4000 STARTED; M=SDOT
HWSS0790I LISTENING ON PORT=5555 STARTED; M=SDOT
HWSC0010I WELCOME TO IMS CONNECT! ID=IMS15HWS
HWSD0290I CONNECTED TO DATASTORE=IVP1 ; M=DREC
HWSC0001I HWS ID=IMS15HWS RACF=Y PSWDMC=R
HWSC0001I UIDCACHE=N UIDAGE=2147483647 SENDALTP= TRANORCH=
HWSC0001I MAXSOC=50 TIMEOUT=5000 TCPIFQ=50
HWSC0001I NUMSOC=2 WARNSOC=80% WARNINC=5%
HWSC0001I RRS=Y STATUS=ACTIVE MULTIRTP= CASCADE=N
HWSC0001I VERSION=V15 IP-ADDRESS=010.001.001.002
HWSC0001I SUPER MEMBER NAME= CM0 ACK TOQ=
HWSC0001I ADAPTER=N MAXCVRT=100 NUMCVRT=0
HWSC0001I MAXLSSSZ=32767
HWSC0001I ODBM AUTO CONNECTION=Y
HWSC0001I ODBM TIMEOUT=6000
HWSC0001I ODBM IMSPLEX MEMBER=IMS15HWS TARGET MEMBER=PLEX1
HWSC0001I DATASTORE=IVP1 STATUS=ACTIVE
HWSC0001I GROUP=OTMAGRP MEMBER=HWSMEM
HWSC0001I TARGET MEMBER=OTMAMEM STATE=AVAIL
HWSC0001I DEFAULT REROUTE NAME=HWS$DEF IMS VERSION=15.1
HWSC0001I RACF APPL NAME=IMSAPPL MULTIRTP= CASCADE=
HWSC0001I OTMA ACCE AGING VALUE=999999 SENDALTP= TRANORCH=
HWSC0001I OTMA ACK TIMEOUT VALUE=120
HWSC0001I OTMA MAX INPUT MESSAGE=5000
HWSC0001I SUPER MEMBER NAME= CM0 ACK TOQ=
HWSC0001I IMSPLEX=PLEX1 STATUS=ACTIVE
HWSC0001I MEMBER=IMS15HWS TARGET=PLEX1
HWSC0001I ODBM=IVP1500D STATUS=REGISTERED ODBMRRS=N
HWSC0001I ALIAS=IVP1 STATUS=ACTIVE
HWSC0001I NO ACTIVE MSC
HWSC0001I NO ACTIVE ISC
HWSC0001I PORT=4000 STATUS=ACTIVE KEEPAV=0 NUMSOC=1 EDIT= TIM
HWSC0001I NO ACTIVE CLIENTS
HWSC0001I PORT=5555D STATUS=ACTIVE KEEPAV=0 NUMSOC=1 EDIT= TIM
HWSC0001I NO ACTIVE CLIENTS
HWSC0001I NO ACTIVE RMTIMSCON
HWSC0001I NO ACTIVE RMTICIS
***** BOTTOM OF DATA *****

```

5. IMS connect can be configured with a file that records activity by providing an internal trace . This is called the **IMS Connect recorder file**.

- To enable this capability, the IMS Connect startup JCL must contain a specification for the sequential file.
- The DD is HWSRCORD in DFSF10.PROCLIB(IMS15HWS)
-

```

File Edit Settings View Communication Actions Window Help
Menu Utilities Compilers Help
BROWSE DFSF10.PROCLIB(IMS15HWS) - 01.00 Line 0000000000 Col 001 080
Command ==> Scroll ==> PAGE
***** Top of Data *****
/* IVP IMS 15.1
/*
/* SKELETON: DFSIXSTG
/*
/* FUNCTION: START IMS CONNECT REGION.
/* *****
/* *****@SCPYRT**
/* LICENSED MATERIALS - PROPERTY OF IBM
/*
/* 5635-A06
/*
/* COPYRIGHT IBM CORP. 1989,1998 ALL RIGHTS RESERVED
/*
/* US GOVERNMENT USERS RESTRICTED RIGHTS - USE, DUPLICATION OR
/* DISCLOSURE RESTRICTED BY GSA ADP SCHEDULE CONTRACT WITH
/* IBM CORP.
/*
/* *****@ECPYRT**
/* IMS15HWS PROC RGN=512M,TME=1440,SOUT=*,
/* BPECFG=BPECFGIV,HWSCFG=HWSCFG0DB
/*
/* HWSREGN EXEC PGM=HWSHWS00,REGION=&RGN,TIME=&TME,
/* PARM='BPECFG=&BPECFG,HWSCFG=&HWSCFG'
/* STEPLIB DD DSN=DFSF10.SDFSRESL,DISP=SHR
/* DD DSN=CEE.SCEERUN,UNIT=SYSALLDA,DISP=SHR
/* DD DSN=SYS1.CSSLIB,UNIT=SYSALLDA,DISP=SHR
/* PROCLIB DD DSN=DFSF10.PROCLIB,DISP=SHR
/* SYSPRINT DD SYSOUT=&SOUT
/* SYSUDUMP DD SYSOUT=&SOUT
/* HWSRCORD DD DSN=DFSF10.HWSRCDR,DISP=SHR
/* PEND
/*
/* HWS01 EXEC IMS15HWS
/*
***** Bottom of Data *****

```

```

Data Set Information
Command ==>
Data Set Name . . . : DFSF10.HWSRCDR
General Data
Management class . . : **None**
Storage class . . . : **None**
Volume serial . . . : C3IMF1
Device type . . . : 3390
Data class . . . : **None**
Organization . . . : PS
Record format . . . : FB
Record length . . . : 1440
Block size . . . : 14400
1st extent cylinders: 1
Secondary cylinders: 0
Data set name type :
Data set encryption : NO
Current Allocation
Allocated cylinders : 1
Allocated extents . : 1
Current Utilization
Used cylinders . . : 0
Used extents . . . : 0
Dates
Creation date . . . : 2018/02/02
Referenced date . . : 2018/02/02
Expiration date . . : ***None***

```


/xxrecorder close

[illegible]

5-e. You can now view the recorder dataset. As you saw in the earlier screenshot, the recorder dataset on this system is DFSF10.HWSRCDR.

```
File Edit Settings View Communication Actions Window
Help
Menu RefList RsfMode Utilities Help

Data Set List Utility

Option ==>
blank Display data set list          P Print data set list
U Display UT00 information            PU Print UT00 information

Enter one or both of the parameters below:
Dsname Level . . . DFSF10.HW5RCDR
Volume serial . . .

Data set list options
Initial View
1. Volume
2. Space
3. Attrib
4. Total

Enter "/" to select option
/ Confirm Data Set Delete
/ Confirm Member Delete
/ Include Additional Qualifiers
/ Display Catalog Name
/ Display Total Tracks
/ Prefix Dsname Level

When the data set list is displayed, enter either:
"/" on the data set list command field for the command prompt pop-up,
an ISPF line command, the name of a TSO command, CLIST, or REXX exec, or
"=" to execute the previous command.
```

Browse the dataset

```

BROWSE DFSF10.HWSRCDR Line 0000000000 Col 001 080
Command ==> Scroll ==> PAGE
*****
.ICONRC. .... I. .... M. .... T. .... F. .... D. .... a. ....
. .... M. .... M. .... M. .... M. .... M. .... M. ....
. .... M. .... M. .... M. .... M. .... M. .... M. ....
. .... M. .... M. .... M. .... M. .... M. .... M. ....
. .... M. .... M. .... M. .... M. .... M. .... M. ....
. .... M. .... M. .... M. .... M. .... M. .... M. ....
*****
***** Bottom of Data *****

```

The ICONRC record shows what IMS connect received

The ICONSN record shows the response from IMS

The following links describe how to read the records in detail

<https://www.ibm.com/docs/en/ims/15.5.0?topic=traces-recorder-log-record-mappings>

<https://www.ibm.com/docs/en/ims/15.5.0?topic=tailoring-jcl-print-ims-connect-recorder-output>

At this point, however, you can get an indication of the messages sent back and forth by scrolling to the right. As you can see, column 642 shows the input to transaction IVTNO. The line under that shows the response.

```

Menu Utilities Compilers Help
BROWSE DFSF10.HWSRCDR Line 00000000001 Col 641 720
Command ==> Scroll ==> PAGE
. ....
.IVTNO DISPLAY Mouse ..... 0004
.IVTNO DISPLAY LAST1 .....
.IVTNO DISPLAY LAST1 FIRST1 .....
*****

```